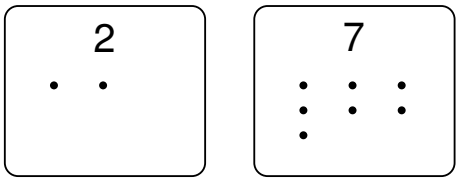
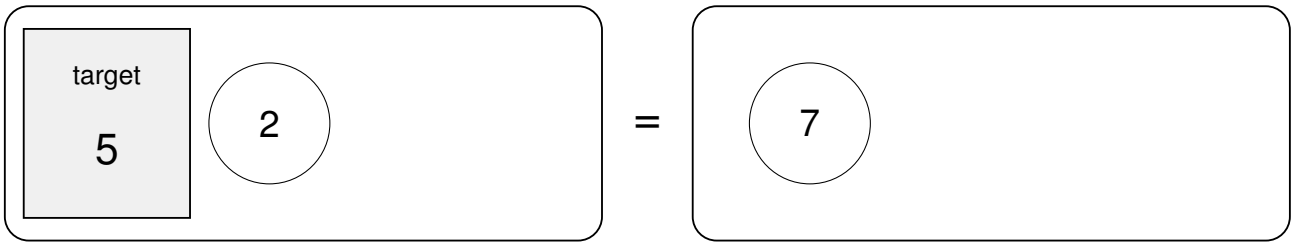
Keep the target on the left. Each weight may go on either pan or stay off. Use each weight at most once. Equal totals balance.

Target 5; weights 2 and 7

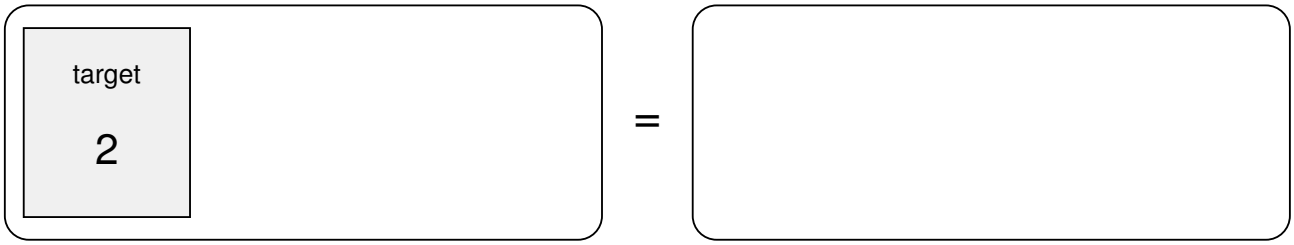


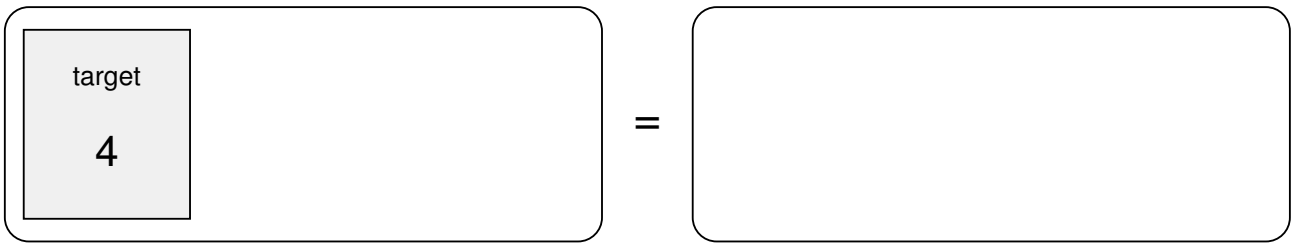

put 2 beside the target




5 + 2 = 7

**Problem 1:** Choose two different whole-number weights that balance every target from 1 through 4. Find every possible kit.



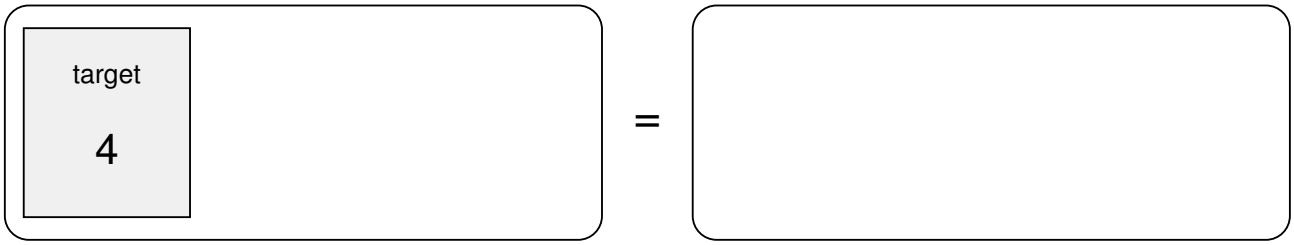
**Problem 2:** Add one weight to 1 and 3. Which choice balances the longest unbroken run of targets starting at 1? Explain why a smaller or larger choice cannot do better.

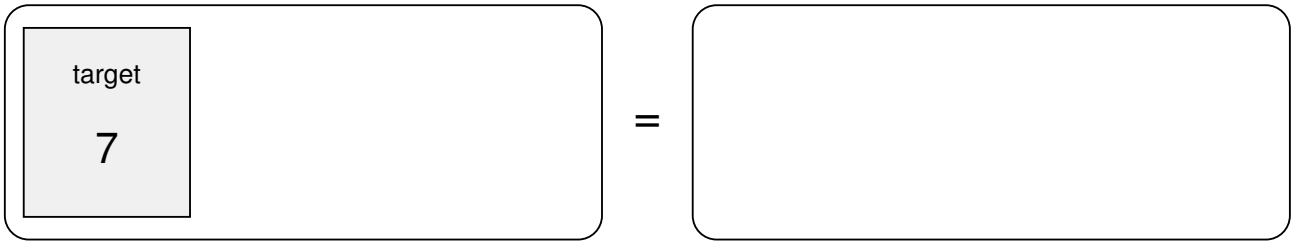
| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| 9 | 10 | 11 | 12 | 13 | 14 | 15 | |

target 5 = 

target 7 = 

**Problem 3:** Can any positive target be balanced in two different ways with weights 1, 3 and 9? Decide for every target this kit can balance.



**Problem 4:** Could any three-weight kit balance every target from 1 through 14? Find one or explain why none can work.

**Problem 5:** Choose three different whole-number weights. How many different positive targets can your kit balance? What is the largest number of different positive targets any three-weight kit can balance, even with gaps? Explain why none can do better.

**Problem 6:** Use the kit 1, 3 and 9. Add one weight so that the new kit balances every target in the longest possible run starting at 1, each in exactly one way. Explain why there are neither gaps nor repeated answers.

**Problem 7:** With five weights, what is the longest run of whole-number targets starting at 1 that any kit could balance? Give a kit that reaches your bound, and explain why it works.